

Sunghwan Yoo

Computer Vision & AI Research and Development

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Education

Gachon University — M.S. student, School of Computing Sep. 2026 – Present
CVIP Lab, Seongnam, South Korea.

Gachon University — B.S., School of Computing Mar. 2021 – Aug. 2026
GPA: 4.47/4.5 overall; 4.5/4.5 major. Graduated first in major with early graduation.

Technical Skills

Tools: Python, PyTorch, OpenCV, YOLO, ONNX.

Areas: Infrared-visible image fusion, image deblurring, object detection, event-based depth reconstruction.

Experience

CVIP Lab, Gachon University Dec. 2023 – Present
M.S. student since Sep. 2026; previously Undergraduate Research Assistant.

- Research image fusion and deblurring; contributed to three KCC conference papers and two journal manuscripts currently under review.

Freelance Computer Vision Engineer — Freemoa Feb. – Mar. 2025

- Developed a YOLO-based model to locate cattle tails and classify raised/lowered states in barn CCTV footage.
- Redesigned labeling criteria, improved training data, analyzed false detections from barn structures, applied OpenCV augmentation, tuned models, and delivered an ONNX model.

Research and Industry Projects

Private 5G Military Communications — LIG Collaboration Mar. 2025 – Present
Researched defense applications and technology trends, planned tactical scenarios, and prepared technical reports using M&S results supplied by a collaborating lab.

DVS-based 3D Depth Reconstruction — NRF Mid-Career Research Sep. 2025 – Present
Research AI algorithms for high-quality 3D depth reconstruction from dynamic vision sensors (DVS).

Advanced GPU Utilization Support Program — MSIT/NIPA Apr. – Jul. 2026
Prepared technical proposals and research reports, analyzed training compute requirements, and planned GPU utilization. Contributed to the application's successful selection.

Selected Publications

- **Infrared-Visible Image Fusion via Heat Conduction-based Frequency Fusion.** First author, KCC 2026.
- **HeatFuse: Heat-Conduction-Based Spectral Decomposition and Cross-Band Modulation for Infrared-Visible Image Fusion.** Under review, *IEEE Trans.*
- **Cross-stage Patch Interaction for Coarse-to-Fine Image Deblurring.** Under review, *MDPI*.

Honors

KCC 2026 Undergraduate Paper Competition — Encouragement Award (Aug. 2026).

National Science and Engineering Excellence Scholarship, Korea Student Aid Foundation (Oct. 2025).